

User as Code: Two-Phase Architecture

Phase 1: Memorizing

(per session, append-only)

Session 1

Session 2

Session 3

...

Extract facts
(thinking LLM)



Append-only Fact List (~1600 facts)

Structure
(thinking LLM)

Phase 2: Structuring

(periodic, from all facts)

Typed Python Code

```
passport = PassportInfo(  
    expiry=date(2025,2,18))  
notes=["Prefers aisle..."]
```

Constraint Execution → ACTIVE_ALERTS



Tier 3: Archive (ChromaDB)

Raw conversation chunks + fact vectors

Multi-Strategy Retrieval

Code + Facts + Archive → Answer

Manifest: Domains + ACTIVE_ALERTS

Always in agent context (~300 tokens)